Bonus Homework 1

Discrete Mathematics by Hongfei Fu, 2023.9.28

Name:Chen Zhuoyan Student ID:523030910037 Class: 2302

1. (Bonus Homework, **0.2pt**) Prove that $\{\neg, \oplus\}$ is not functional complete.
Hint: First prove from the commutativity, associativity of $\oplus$ and the law $\neg(p \oplus q) \equiv (\neg p) \oplus q$ that any propositional formula that uses only $\neg$ and $\oplus$ can be equivalently transformed into a group-$\oplus$ of propositional variables p and their negations $\neg p$. Then eliminate duplicate propositional variables by using $p \oplus (\neg p) \equiv \mathbf{T}$ and $p \oplus p \equiv \mathbf{F}$. Finally, it should be obvious that the truth of such a propositional formula depends only on the number of propositional variables taking the truth value $\mathbf{T}$, so that conjunction and disjunction cannot be expressed by such propositional formula.

Answer Area:

First, we prove the commutativity of $\oplus$:

p	q	$p \oplus q$	$q \oplus p$
T	T	F	F
T	F	T	T
F	T	T	T
F	F	F	F

$\therefore p \oplus q \equiv q \oplus p$.

Next, we prove the associativity of $\oplus$:

p	q	r	$(p \oplus q) \oplus r$	$p \oplus (q \oplus r)$
T	T	T	T	T
T	T	F	F	F
T	F	T	F	F
T	F	F	T	T
F	T	T	F	F
F	T	F	T	T
F	F	T	T	T
F	F	F	F	F

$\therefore (p \oplus q) \oplus r \equiv p \oplus (q \oplus r)$.

Then, we prove $\neg(p \oplus q) \equiv (\neg p) \oplus q$:

p	q	$\neg(p \oplus q)$	$(\neg p) \oplus q$
T	T	T	T
T	F	F	F
F	T	F	F
F	F	T	T

$\therefore \neg(p \oplus q) \equiv (\neg p) \oplus q$.

For any propositional formula that uses only $\neg$ and $\oplus$, we can equivalently transform it into a group-$\oplus$ of propositional variables and their negations.

$\therefore$ By using commutativity and associativity of $\oplus$,we can combine p and $\neg p$, or p and p.Then we can equivalently transform them into T or F by using $p \oplus \neg p \equiv T$ and $p \oplus p \equiv T$.

After that, the propositional formula consists of T, F and propositional variables or their negations (without duplicate propositional variables),linked by $\oplus$.

Then,we can eliminate T and F by using $T \oplus p \equiv \neg p$ and $F \oplus p \equiv p$,transforming the formula into a group-$\oplus$ of propositional variables or their negations.

When we assign these propositional variables truth values, we can use $F \oplus F \equiv F$ to transform the formula into a group-$\oplus$ of Ts and just one(or even no) F.So the truth value of such a propositional formula depends only on the number of propositional variables taking the truth

value T.
$\therefore$ The disjunction $p \vee q$ can only be expressed as $p \oplus q$, $\neg p \oplus q$, $p \oplus \neg q$ or $\neg p \oplus \neg q$.Now we prove that none of these four propositional formulas is logically equivalent to $p \vee q$.

p	q	$p \vee q$	$p \oplus q$	$\neg p \oplus q$	$p \oplus \neg q$	$\neg p \oplus \neg q$
T	T	T	F	T	T	F
T	F	T	T	F	F	T
F	T	T	T	F	F	T
F	F	F	F	T	T	F

$\therefore$ Disjunction cannot be expressed by such propositional formula.It can also be proved by the similar reason that conjunction cannot be expressed by such propositional formula.
$\therefore$ $\{\neg, \oplus\}$ is not functional complete.